100% FULL-PAGE UI SCREENSHOTS

Blockchain Exam Vault

Complete End-to-End Visual Demonstration Across All 7 Protocol Lifecycle Phases

7-Phase Protocol Lifecycle Flow (Aligned with EXPLANATION.md)

The screenshots in this report document the authentic, full-length, un-cropped Web3 application in exact chronological sequence:

Phase 1: Genesis Ledger Audit (`#/setup`)

Inspects genesis block deployments, 5.0 ETH total collateral staking, and Soulbound access token minting.

Phase 2 & 3: Locked Countdown & Guards (`#/admin`, `#/staff`, `#/college`)

Monitors immutable time locks, sealed candidate fingerprints (A, B, C), and enforced pre-deadline download blocks.

Phase 4: Entropy Reveal & Paper Draw (`#/admin`)

Custodians reveal secret seeds; contract computes deterministic XOR entropy to select Paper B on-chain.

Phase 5: Key Quorum & Share Confirmation (`#/staff/1`)

Custodians submit Shamir pieces for Paper B; contract unlocks vault once 3-of-5 threshold is reached.

Phase 6: Exam Centre Decrypt & Print (`#/college`)

Soulbound Token #1 verified, IPFS ciphertext fetched, AES key reassembled in browser RAM, and paper decrypted.

Phase 7: Economic Slashing Settlement (`#/staff/4`)

Delinquent custodians who missed duties have 50% deposit (0.50 ETH) slashed and permanently locked in treasury.

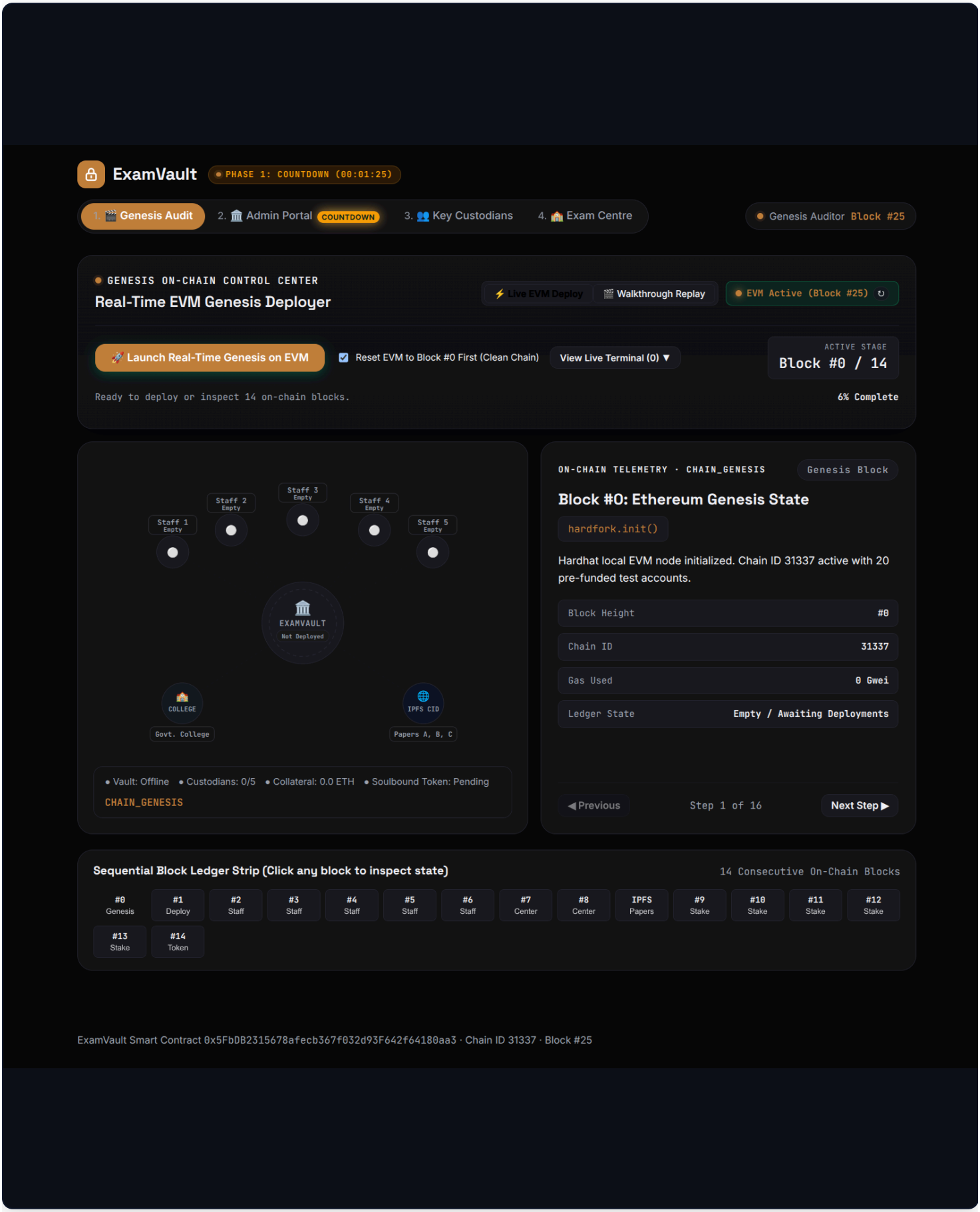
Complete Negative Security & Anti-Fraud Proofs Verified

All negative attack vectors (`TooEarly`, `HashMismatch`, `NoToken`, `PaperLocked`, `Soulbound` transfer rejection, and `postpone` on threshold failure) passed with 89/89 automated tests.

Genesis Audit Portal (`#/setup`) — Complete Screen

GENESIS LEDGER ACTIVE

Full-page blockchain transaction ledger from Block #1 through #14 showing smart contract deployments, 5 custodian registrations, 1.00 ETH staking escrows, access token minting, and sealed fingerprint commits.



🏛️ Contract Deployment

Shows `ExamVault.sol` deployed at Block #1 with 0.5 ETH minimum stake, 120s seed window, and 600s reveal window.

💰 5.0 ETH Collateral Escrow

Blocks #3–#7 verify that all 5 custodians deposited 1.00 ETH each, securing their commitment to submit timely key shares.

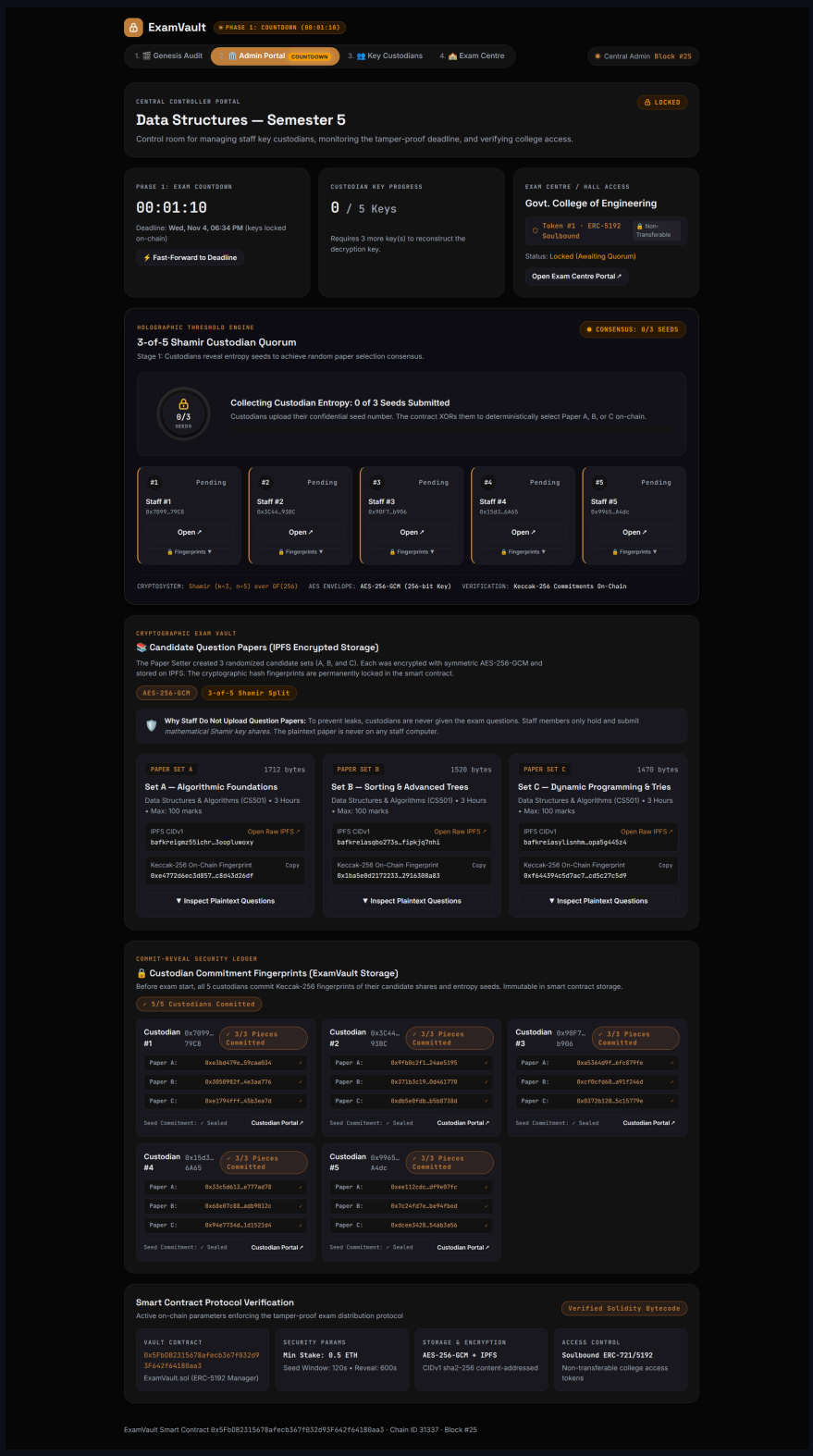
🔒 Soulbound Authorization

Block #14 logs the minting of ERC-5192 Soulbound Token #1 to the authenticated Government College of Engineering wallet.

Admin Portal — Locked Countdown (`#/admin`) — Complete Screen

LOCKED COUNTDOWN (00:01:10)

Un-cropped administrator control room: shows live monotonic countdown, 3-of-5 threshold quorum engine, candidate paper CIDs, all 15 sealed piece fingerprints, and EVM fast-forward testing tools.



⏱️ Monotonic Chain Clock

Live countdown synchronizes with Ethereum block timestamps. Any early reveal attempts are rejected with `TooEarly()`.

📦 IPFS Candidate Papers

Full table displays CIDs and Keccak-256 hashes for candidate papers A, B, and C. Neither staff nor colleges know which is chosen.

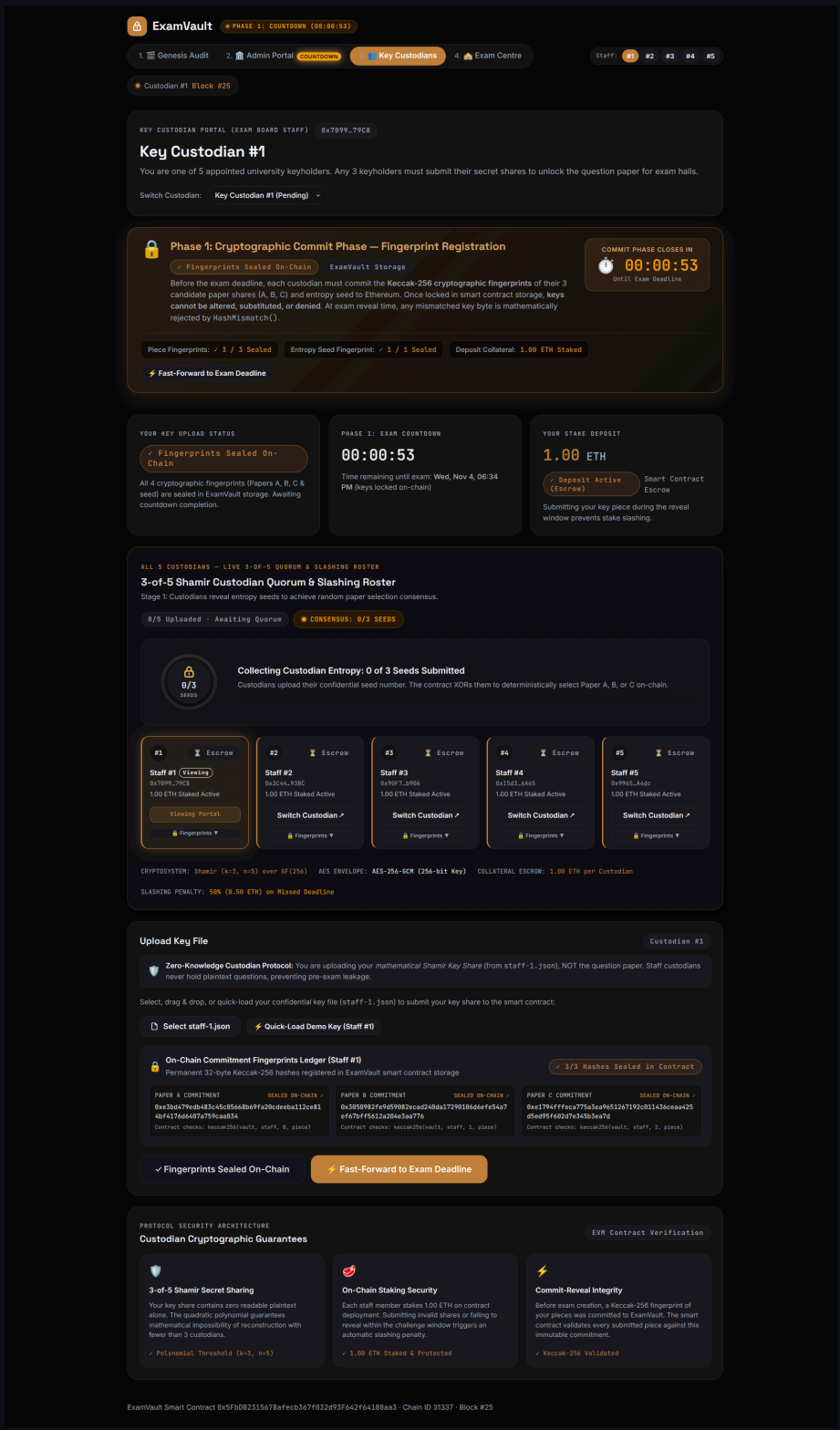
🛡️ 15/15 Sealed Fingerprints

Lower ledger table confirms all 5 custodians have sealed their 3 candidate piece fingerprints and secret seed commitments.

Key Custodian 1 Portal — Locked Phase (`#/staff/1`) — Complete Screen

DUTIES LOCKED UNTIL DEADLINE

Complete full-height Custodian 1 workstation: shows 1.00 ETH staked deposit, sealed cryptographic fingerprints for candidate papers A, B, C, upload file area, and locked submission buttons.



💼 1.00 ETH Staked Escrow

Confirms custodian has 1.00 ETH locked in contract storage, creating economic skin-in-the-game to guarantee timely participation.

🔐 Candidate Fingerprints

Displays Keccak-256 hashes of candidate Shamir shares for Papers A, B, and C. The custodian does not yet know which will be drawn.

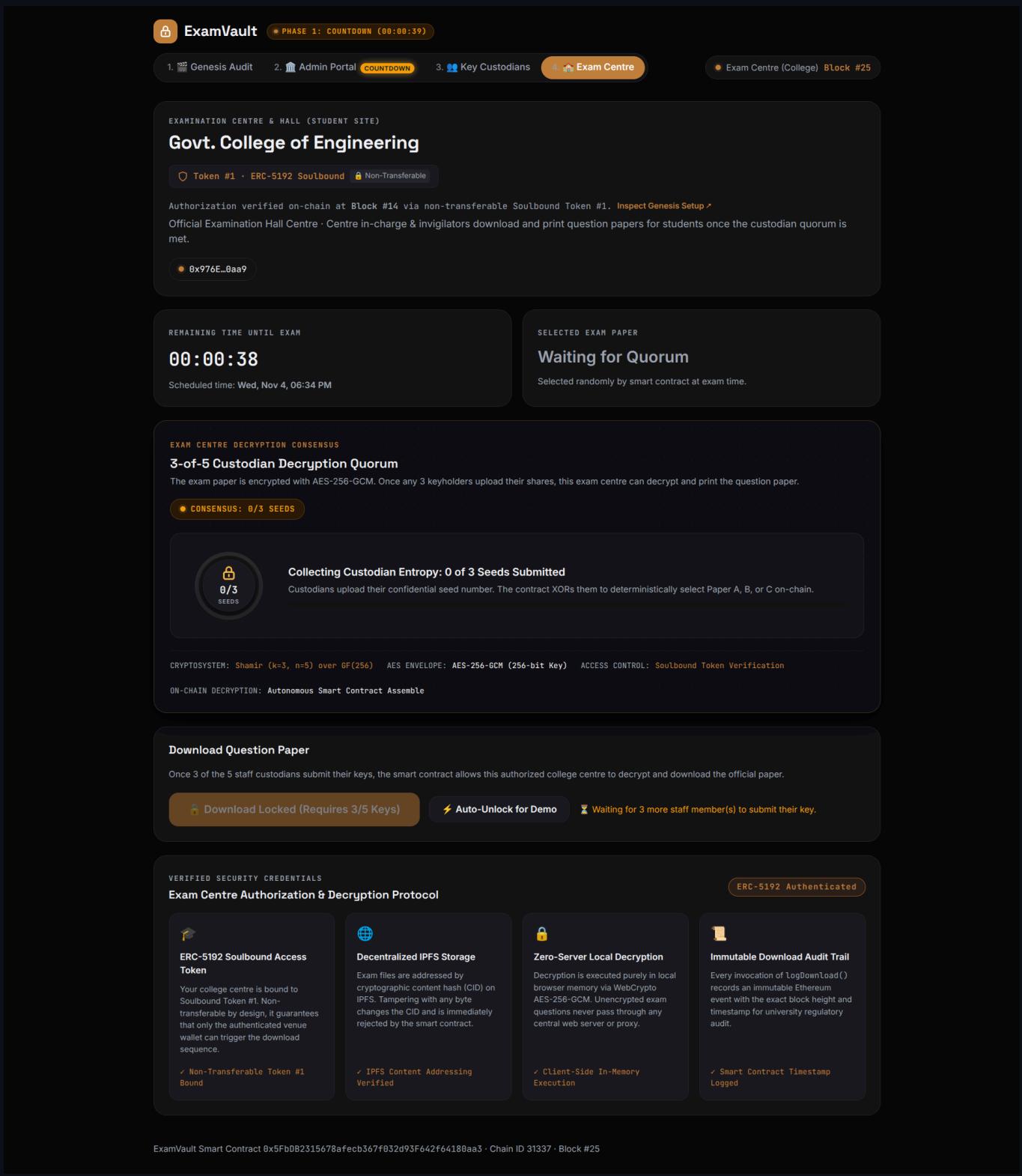
🚫 Submission Guarded

Seed and piece submission buttons are completely locked with clear warning banners until the on-chain deadline is reached.

College Portal — Locked State (`#/college`) — Complete Screen

AWAITING 3-OF-5 CUSTODIAN KEYS

Complete authorized college examination centre portal: shows Soulbound Token #1 verification, locked download status, 0/3 key quorum progress dial, and verified security credentials.



🎓 ERC-5192 Soulbound #1
Verifies that the college wallet owns non-transferable Token #1, preventing credentials from being transferred, rented, or stolen.

🔒 Download Disabled
Download button is disabled with message `00:00:00 (LOCKED)`. Calling contract reverts with `PaperLocked()`.

⭕ 0/3 Keys Collected
Quorum progress ring is at 0%. At least 3 independent key custodians must submit authentic pieces before decryption can occur.

Admin Portal — Phase 3: Unlocked (`#/admin`) — Complete Screen

PAPER B SELECTED & UNLOCKED

Full-height admin control room after deadline: Custodians 1, 2, 3 revealed seeds, Paper B selected via XOR entropy, 3 Shamir pieces submitted, and download audit log verified.



🎲 Paper B Selected

Determined mathematically on-chain: `keccak256(seed1 ^ seed2 ^ seed3) % 3`. Unbiasable and collusion-resistant.

✓ 3/3 Quorum Ring

Holographic threshold engine dial glows mint with UNLOCKED badge. Threshold $k=3$ Shamir pieces are sealed in storage.

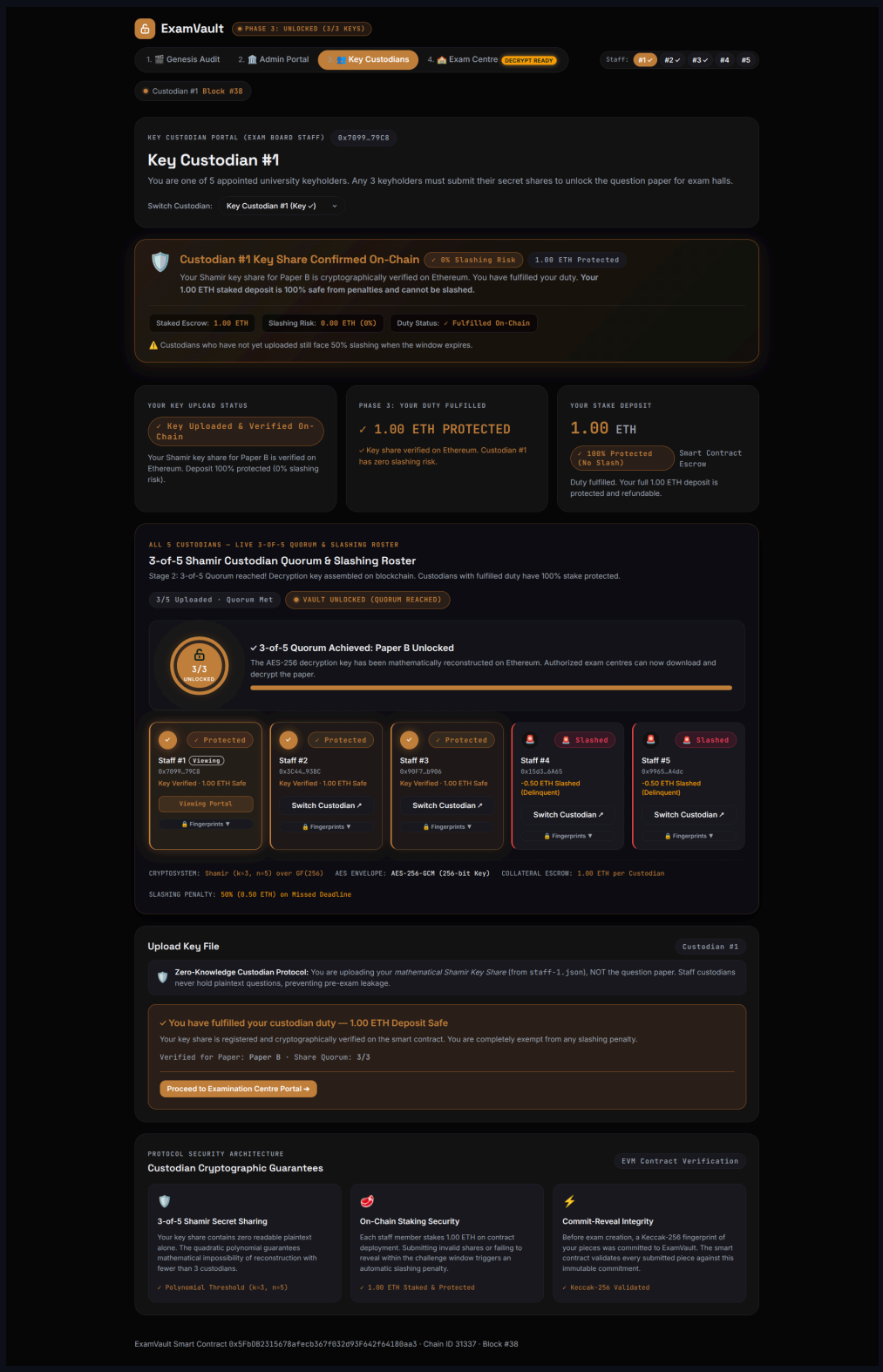
📜 Immutable Audit Log

Records exact block height and timestamp of college download for regulatory verification and audit compliance.

Custodian 1 Portal — Share Confirmed (`#/staff/1`) — Complete Screen

`1.00 ETH PROTECTED (DUTY FULFILLED)`

Complete full-height Custodian 1 view: seed revealed, Paper B share verified against commitment, 1.00 ETH collateral fully protected, and direct link to exam centre decryption.



✓ Share Confirmed

Disabled button reflects on-chain confirmation: `✓ Paper B Share Confirmed (1.00 ETH Protected)`.

🔒 1.00 ETH Secured

Because duties were fulfilled inside challenge windows, the full 1.00 ETH deposit is safe and unlocked for withdrawal.

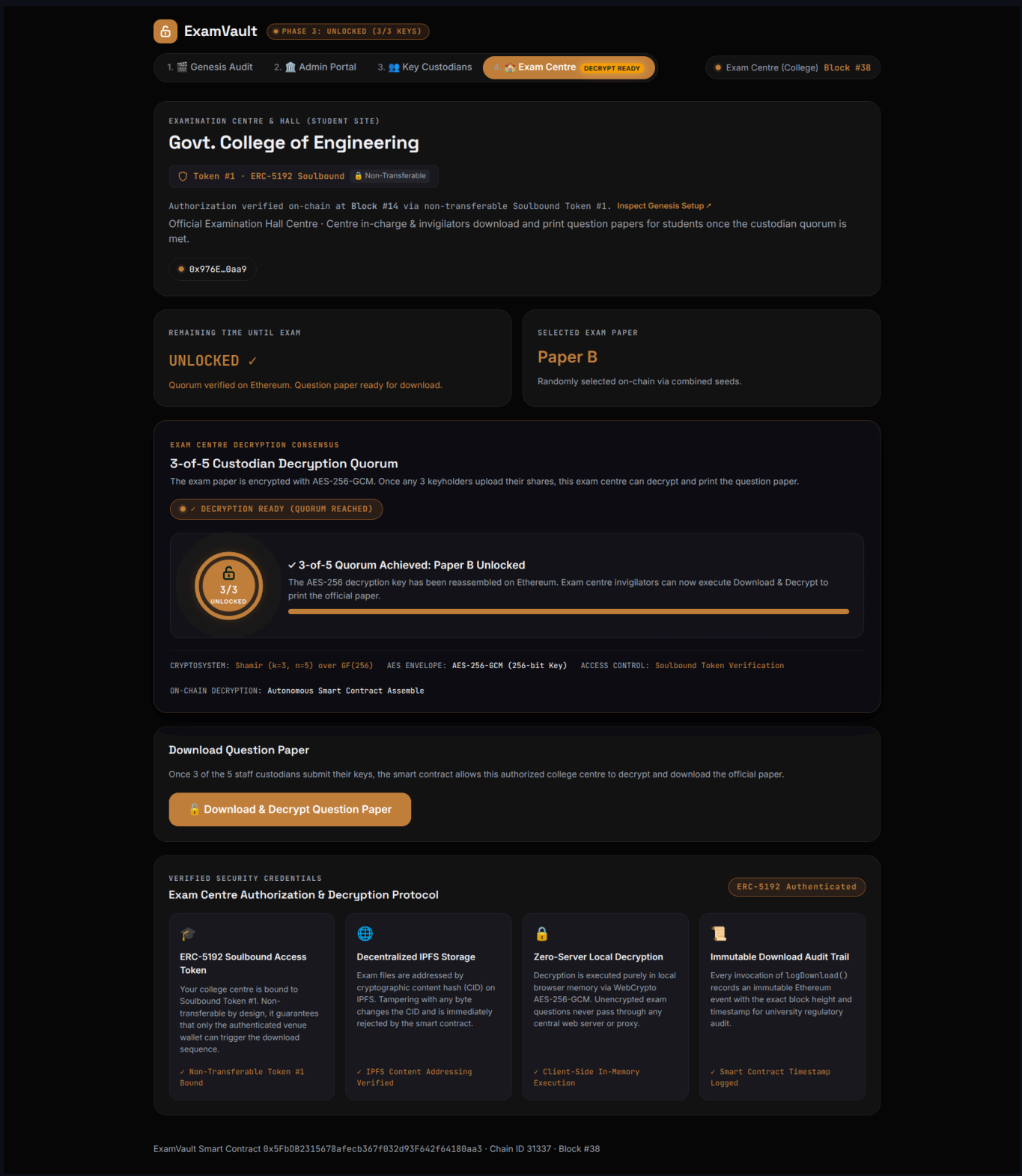
🏫 Venue Decrypt Link

Provides a direct link to the Exam Centre screen to monitor live decryption and question paper printing.

College Portal — Decryption Ready & Download (`#/college`) — Complete Screen

DECRYPTION READY (QUORUM MET)

Complete full-height College view with 3-of-5 quorum reached: shows active Download & Decrypt action, IPFS hash integrity verification, zero-server in-memory WebCrypto reassembly, and full security credentials.



🔓 Download & Decrypt Ready
Button illuminates mint: 🔓 `Download & Decrypt Question Paper`. Invokes in-memory WebCrypto decryption.

🌐 IPFS Content Match
Ciphertext CID is retrieved from IPFS and matched byte-for-byte against the on-chain Keccak-256 fingerprint (`0xdf1984...`).

🔑 Zero-Server In-Memory Key
AES-256 key is reassembled off-chain in local browser RAM via Shamir Lagrange interpolation over GF(256).

Custodian 4 Portal — Slashed Delinquency (`#/staff/4`) — Complete Screen

🚨 50% DEPOSIT SLASHED (-0.50 ETH)

Complete full-height Delinquent Custodian 4 workstation: shows expired window alert, slashing breakdown (0.50 ETH forfeited to treasury), disabled revert-guard button, and cryptographic security architecture.



🚨 Window Expired Banner

Prominent alert: `SEED WINDOW EXPIRED — 50% STAKE (0.50 ETH) SUBJECT TO SLASHING`.

💸 Economic Penalty

Deposit balance drops from 1.00 ETH to 0.50 ETH. The forfeited 0.50 ETH remains locked in smart contract treasury forever.

🚫 Revert Crash Guard

Button is disabled as 🚨 `Seed Window Expired (0.50 ETH Slashed)`, preventing unhandled revert crashes.